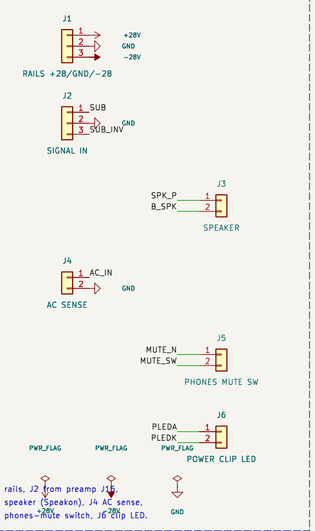
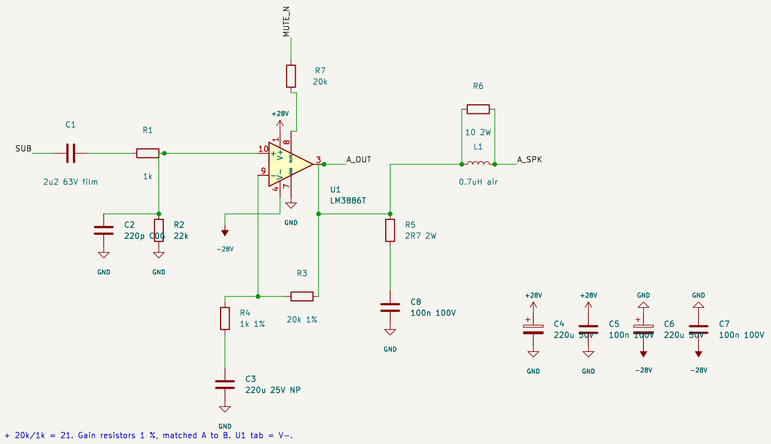
Connectors



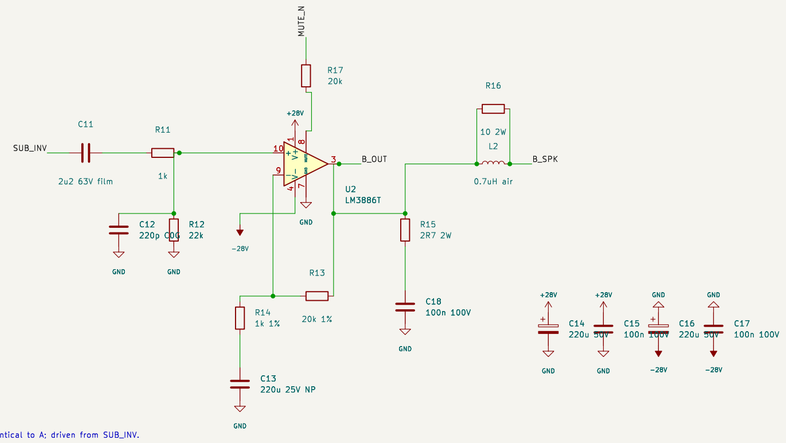
J1 rails, J2 from preamp J16,
J3 speaker (Speakon), J4 AC sense,
J5 phones-mute switch, J6 clip LED.

LM3886 A (non-inverted half)



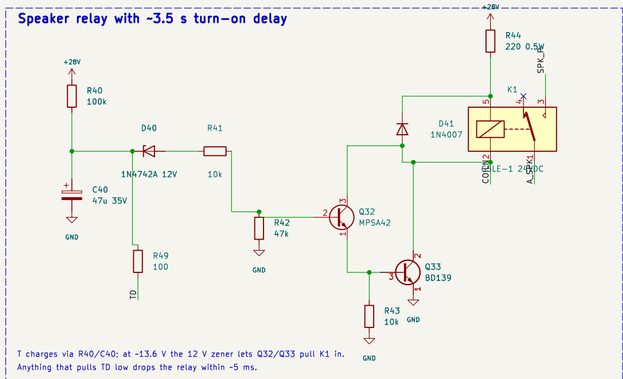
Gain 1 + 20k/1k = 21. Gain resistors 1 %, matched A to B. U1 tab = V−.

LM3886 B (inverted half)



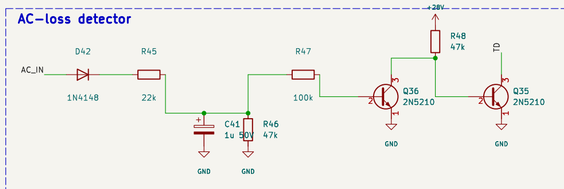
Identical to A: driven from SUB_INV.

Speaker relay with ~3.5 s turn-on delay



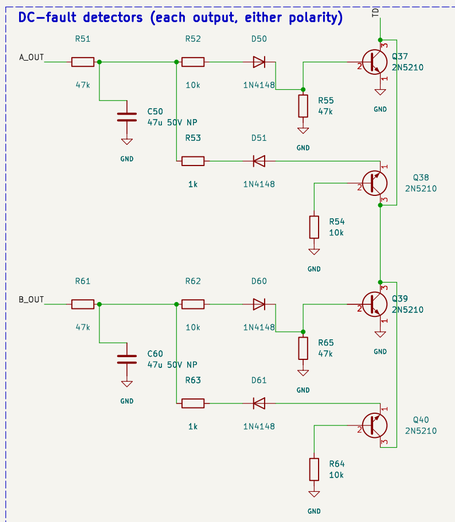
T charges via R40/C40; at ~13.6 V the 12 V zener lets Q32/Q33 pull K1 in.
Anything that pulls TD low drops the relay within ~5 ms.

AC-loss detector



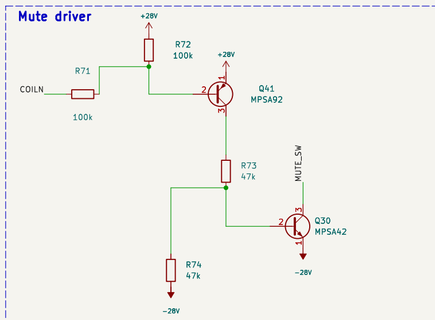
While AC is present Q36 holds Q35 off. Mains off: C41 decays in ~30 ms,
Q35 dumps C40, relay opens, chips mute.

DC-fault detectors (each output, either polarity)



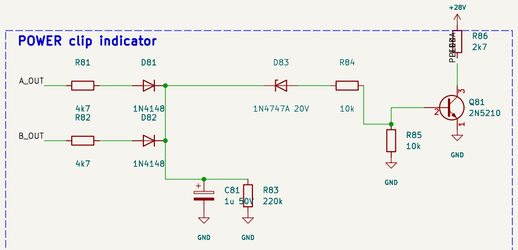
Output low-passed by 47k/47uF NP. > +1.3 V turns on Q37/Q39, < −1.3 V turns on Q38/Q40;
both dump TD. A 25 Vpk 30 Hz bass note leaves only 66 mV: no false trip.

Mute driver



Relay energised (COILN low) -> Q41 -> Q30 on ->
~1.3 mA out of each MUTE pin = play.

POWER clip indicator



Peak detector on both bridge halves; lights at ~21 V peak.

J5 goes to the phones jack NC switch (opens = speaker muted). Fit a jumper if unused.
Heatsink <= 0.4 C/W (or <= 0.7 C/W + fan). LM3886T tab is V−: insulate it.
Rails +/−28 V from the PSU board with heavy wire. Star ground at the PSU reservoir caps.
BRIDGED OUTPUT: neither speaker terminal is ground. Use a Speakon NL4 (1+ = SPK+, 1− = SPK−).

Sheet: /		
File: poweramp.kicad_sch		
Title: Bass amp power amplifier: 2 x LM3886T bridged (BTL), protection		
Size: A2	Date:	**Rev: 1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1